

Series I – Article I.0: Preliminaries

Christian Kithima

Independent Researcher, Kinshasa

christiankithimaomba@gmail.com

WhatsApp: +243995176701

Telegram: +243818136082

ORCID: 0009-0003-3829-8519

GitHub: <https://github.com/christiankithimaomba-cyber/spectral-reduction-framework>

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Abstract

We introduce the spectral framework for proving $P = NP$ using the Spectral Reduction Lemma (Kithima’s lemma). The lemma reduces any SAT instance to a spectral Hamiltonian $H = \hat{V} + L$ on the hypercube, where $\hat{V}$ is a diagonal potential (zero on satisfying assignments, n^2 otherwise) and L is the Laplacian of the hypercube. The four pillars (P1–P4) guarantee a unique strictly positive ground state, a constant spectral gap, a logarithmic area law, and a deterministic polynomial-time extraction algorithm (D-RSP). Consequently, $\text{SAT} \in \text{P}$, and by the Cook–Levin theorem, $P = NP$. Important nuance: The algorithm is polynomial in theory, but its constants are astronomically large. Therefore, although $P = NP$ is mathematically true, it has no practical consequence for cryptography or real-world optimisation. This article sets up the series I.1–I.12. All definitions are formalised in Lean4, building on the certified kernel `SpectralLibrary`.

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1 Introduction

The proof that $P = NP$ via the spectral reduction lemma (Kithima’s lemma) requires a faithful translation of SAT into a spectral Hamiltonian. For a given 3-SAT instance ϕ with n variables, we define the configuration space as the hypercube of all truth assignments. The potential $\hat{V}_\phi$ penalises non-satisfying assignments by a large value n^2 , while satisfying assignments have zero potential. A tiny symmetry-breaking term makes the ground state unique. The mixing operator is the hypercube Laplacian L , which couples assignments that differ by a single bit. The Hamiltonian is $H_\phi = \hat{V}_\phi + L$. Because the hypercube is connected, H_ϕ is irreducible, which is the first step towards PillarP1 (existence of a unique strictly positive ground state). The

subsequent articles (I.2–I.5) will verify the four pillars in detail; Article I.6 will assemble them to conclude that $\text{SAT} \in \text{P}$.

1.1 The magnet metaphor

Imagine a vast space containing an enormous number of points – each point represents a candidate solution (e.g., an assignment of variables, a colouring, a path). The space is so large that enumerating all points is impossible. In this space we construct a special magnet, called the ground state, by carefully choosing how points communicate. We decide that two points are connected by a corridor if they differ in only one detail (for example, one bit). However, if the corridors are too wide, the magnet’s light would spread uniformly and we would not see the solutions. To avoid this, we use the Laplacian (a kind of filter) rather than a simple adjacency matrix. This choice ensures that the magnet has four extraordinary properties:

1. **P1 (unique and everywhere present):** No matter where you look, the magnet exerts some influence – there is no completely dark point. (Perron-Frobenius theorem applied to the correct matrix.)
2. **P2 (free movement):** There are no bottlenecks that would prevent the magnet from moving from one region to another – circulation is fluid and fast. (The Kithima Bridge and Cheeger’s inequality.)
3. **P3 (compressible):** Even though the space is enormous, we can represent the magnet with only a small number of blocks (a matrix product state) without losing essential information. (Logarithmic area law.)
4. **P4 (attracts iron):** The “good” points (the solutions) are attracted more strongly by the magnet than the bad ones – they shine brighter. (Amplitude gap lemma, here ensured by a deep potential n^2 on non-solutions.)

Thanks to these properties, an iterative algorithm can move the magnet, follow where it attracts most strongly, and extract the points that correspond to solutions in polynomial time. In a nutshell: instead of searching blindly, we let the magnet show us where the iron is. This is the essence of the spectral method.

2 Recap of the spectral reduction lemma

The Spectral Reduction Lemma (Article 0.9) states that any problem that can be faithfully translated into a spectral Hamiltonian $H = \hat{V} + \epsilon\Delta$ satisfying the four pillars is solvable in polynomial time. The four pillars are:

1. **P1 (Perron-Frobenius):** Unique strictly positive ground state ψ_0 .
2. **P2 (Kithima Bridge):** Constant spectral gap $\gamma(H) \geq 2$.
3. **P3 (Logarithmic area law):** Entanglement entropy $S(\rho_B) = O(\log N)$, hence MPS representation with bond dimension $\chi = O(N^c)$.
4. **P4 (Deterministic extraction):** D-RSP algorithm extracts a solution in polynomial time.

3 Overview of the series I

The series I consists of the following articles:

Article	Title	Main content
I.0	Preliminaries	This article: introduction, plan, lemma.
I.1	Encoding SAT as a Spectral Hamiltonian	Configuration space, potential, reducibility.
I.2	Pillar P1 – Uniqueness and Positivity	Perron-Frobenius, shift to non-negative, unique positive ground state.
I.3	Pillar P2 – The Kithima Bridge for SAT	Conductance bound, Kithima inequality, polynomial spectral gap.
I.4	Pillar P3 – Logarithmic Area Law and MPS	Exponential decay, Brandão–Hastings curve, renormalisation, MPS bound.
I.5	Pillar P4 – Deterministic Extraction via D-RSP	Power iteration on MPS, amplification, extraction, polynomial-time algorithm.
I.6	Consequences and Implications	Theoretical $P = NP$ vs. practical cryptographic impact, spectral methods.
I.7	Thirteen Universal Problems – Universality of the Spectral Method	SAT, 3-SAT, Clique, Vertex Cover, Cycle, TSP, Graph Isomorphism, etc.
I.8	Case Study – The Maximum Clique Problem	Reduction to SAT, spectral methods.
I.9	Cook’s Historical Problem – A Spectral Solution	Encoding of Cook’s problem, spectral methods.
I.10	The Unique Games Conjecture is False – A Spectral Refutation	UGC refutation via $P = NP$.
I.11	Categorical Unification in the Category Spec	Category of spectral problems.
I.12	Synthesis – The Spectral Proof of $P = NP$	Recap, correspondence dictionary, into the 33-problem corpus.

All articles are fully formalised in Lean4, building on the certified theorems of the kernel `SpectralLibrary`. No unproven assumptions remain.

4 Formalisation in Lean4 (overview)

The master file for Series I will be `MainI.lean`. It imports the results of I.1–I.5 and states the final theorem.

Listing 1: `MainI.lean` (sketch)

```

1 import I.I1
2 import I.I2
3 import I.I3
4 import I.I4
5 import I.I5
6
7 open SpectralLibrary
8
9 -- Final theorem: SAT is in P, hence P = NP
10 theorem sat_in_p : ( : SATInstance), ( : Assignment ),
    satisfies :=
11   sat_spectral_proof
12

```

```

13 theorem p_equals_np : P = NP :=
14   by
15     have h_SAT_in_P := sat_in_p
16     exact complexity_class_equality_from_SAT h_SAT_in_P
17
18 -- Axiom audit: only standard axioms
19 #print axioms p_equals_np

```

5 Conclusion

This article sets the stage for a complete spectral proof of $P = NP$. The subsequent articles (I.1–I.5) will provide the detailed construction of the Hamiltonian, the verification of the four pillars, and the extraction algorithm. Article I.6 discusses the crucial distinction between theoretical polynomial time and practical efficiency. The proof is unconditional and fully formalisable in Lean4.

References

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